Matrix discrepancy and optimization

# The sharp universal Spencer discrepancy constant

**Difficulty:** hard **Importance:** interesting to the community

**Provenance:** source-stated extremal-value problem.

**Status:** no determination of the constant found in screening on 2026-09-08.

For each positive integer $n$, define

$$d_n = \frac{1}{\sqrt{n}} \max_{A\in\{-1,1\}^{n\times n}} \min_{x\in\{-1,1\}^n} \|Ax\|_\infty.$$

Determine the exact value of

$$C_* = \sup_{n\geq 1} d_n.$$

Thus $C_*$ is the smallest real constant for which every square sign matrix $A$ admits a column signing $x$ with $\|Ax\|_\infty \leq C_* \sqrt{n}$.

The question concerns the sharp worst-case error in simultaneous signed sums. In matrix form it asks how far a square sign matrix can keep all sign vectors from the coordinate cube of radius $C\sqrt{n}$.

The source also poses $\limsup_{n\to\infty} d_n > 1$ as a related asymptotic conjecture. It is recorded here as context rather than counted as another entry.

## References

1. A. S. Bandeira, A. Kireeva, A. Maillard, and A. Rödder, *Randomstrasse101: Open Problems of 2024*, arXiv:2504.20539 (2025), entry 6, Open Problem 11, Conjecture 12, and the subsequent "Updates" paragraph. Paper.
2. A. S. Bandeira, *Did just a couple of deviations suffice all along? (problems 10–14)*, Randomstrasse101 (2024), the original statement of Open Problem 11. Author's research post.
3. J. Spencer, *Ten Lectures on the Probabilistic Method*, second edition, SIAM (1994), Chapter 10, "Six Standard Deviations Suffice," pp. 75–80, the classical universal-constant upper bound. Chapter.

## Status check — 2026-09-08

checked the current arXiv record, still 2504.20539v1 (2025-04-29), the source's update, and searches "Spencer sharp constant discrepancy" and "Spencer limsup discrepancy 2026". The update refutes the stronger odd Sylvester–Hadamard claim in Conjecture 13; that claim is excluded. No resolution of Open Problem 11 or Conjecture 12 was found. Recent matrix Spencer results concern a different matrix-valued discrepancy problem.